

SQL Window Function Analytics Report

Objective

Performed exploratory and business-focused analysis on retail transaction data using PostgreSQL and Power BI to identify sales trends, customer behavior, and profitability insights.

Tools & Technologies

PostgreSQL | SQL | pgAdmin | Power BI | CTEs | Window Functions

Key Highlights

1. Analyzed 9,994 retail transactions to evaluate sales, profit, and regional business performance using advanced SQL queries and aggregations.
 2. Applied SQL concepts such as CTEs, joins, window functions, ranking functions, and cohort analysis to uncover customer and product-level insights.
 3. Identified top-performing products, profitable customer segments, and Pareto 80/20 revenue contributors to support business decision-making.
 4. Performed time-series analysis including Month-over-Month (MoM) and Year-over-Year (YoY) growth tracking to monitor sales trends and profitability patterns.
 5. Planned and prepared a Power BI dashboard for interactive visualization of KPIs, customer segmentation, regional performance, retention analysis, and sales trends.
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Query 1 — Total Revenue by Region

What this query does

Calculates the total sales revenue and total profit generated from each region by aggregating transaction-level data.

Business Insight

Helps identify:

- Highest revenue-generating regions
- Most profitable business areas
- Regional performance comparison for strategic decision-making

Query 2 — Top 10 Most Profitable Products

What this query does

Identifies the top 10 products generating the highest overall profit by calculating total sales and profit for each product and category.

Business Insight

Helps identify:

- Most profitable products in the business
- High-performing product categories
- Products contributing maximum profit to overall revenue

Query 3 — Revenue by Category and Sub-Category

What this query does

Calculates total revenue, total profit, and profit margin percentage for each product category and sub-category.

Business Insight

Helps identify:

- Most profitable product categories
- Sub-categories with high or low profit margins
- Product segments driving business growth and profitability

Query 4 — Rank Customers by Revenue

What this query does

Ranks customers based on their total sales contribution using the `DENSE_RANK()` window function while also grouping them by customer segment and region.

Business Insight

Helps identify:

- Highest revenue-generating customers
- Valuable customer segments
- Regional distribution of top customers for targeted business strategies

Query 5 — Running Total of Sales by Date

What this query does

Calculates daily sales revenue and generates a cumulative running total of sales over time using window functions and a Common Table Expression (CTE).

Business Insight

Helps identify:

- Overall sales growth trend over time
- Revenue accumulation patterns
- Business performance progression across different periods

Query 6 — Month over Month Revenue Growth

What this query does

Calculates monthly revenue and compares it with the previous month using the `LAG()` window function to measure month-over-month growth percentage.

Business Insight

Helps identify:

- Monthly business growth trends
- Revenue increase or decline patterns
- Seasonal sales fluctuations and performance changes over time

Query 7 — Year over Year Revenue by Region

What this query does

Calculates yearly revenue for each region and compares it with the previous year using the `LAG()` window function and `PARTITION BY` clause to measure year-over-year growth percentage.

Business Insight

Helps identify:

- Regional growth and decline trends over multiple years
- High-performing and underperforming regions
- Long-term business expansion patterns across regions

Query 8 — Top 3 Customers per Region

What this query does

Identifies the top 3 highest revenue-generating customers in each region using the `ROW_NUMBER()` window function with regional partitioning.

Business Insight

Helps identify:

- Most valuable customers in every region
- Regional customer contribution patterns
- High-priority customers for retention and targeted marketing strategies

Query 9 — Customer Segmentation by Revenue

What this query does

Segments customers into High Value, Mid Value, and Low Value categories based on their total sales revenue using CTEs and a `CASE` statement.

Business Insight

Helps identify:

- High-value customers contributing maximum revenue
- Customer purchasing behavior across segments
- Opportunities for targeted marketing and customer retention strategies

Query 10 — Cohort Retention Analysis

What this query does

Tracks customer retention over time by grouping customers into cohorts based on their first purchase month and calculating retention rates for subsequent months.

Business Insight

Helps identify:

- Customer retention and repeat purchase behavior
- Long-term customer engagement trends
- Effectiveness of customer loyalty and retention strategies

Query 11 — Discount Impact on Profit

What this query does

Classifies orders into different discount tiers using a **CASE** statement and analyzes their impact on revenue, profit, and profit margin percentage.

Business Insight

Helps identify:

- Impact of discounts on business profitability
- Optimal discount ranges for maintaining healthy margins
- Whether higher discounts increase sales at the cost of profit

Query 12 — Pareto 80/20 Analysis

What this query does

Performs Pareto analysis by calculating cumulative customer revenue contribution and identifying the group of customers responsible for approximately 80% of total revenue.

Business Insight

Helps identify:

- High-value customers contributing the majority of revenue
- Revenue concentration patterns across the customer base
- Priority customers for retention and targeted business strategies

Key Findings from SQL Analysis

- Highest revenue region → **West \$739813.71**
- Most profitable product → **Canon imageCLASS 2200 Advanced Copier + \$X 25199.94**
- Any negative margin sub-category → **Furniture(Tables) + -8.53% margin**
- Top customer → **Tamara Chand + \$18437.14**
- - High value customers → **119 total, driving \$933255.45 revenue**
- Cohort retention month 3 → **6.06%**
- High discount margin vs no discount → **37.29% vs 29.56%**
- Pareto → **top 300 customers = 80% revenue**